

# Note (Revision 5.1): Controlled Spatial-Convergence Study of the Drain-Localized Impact-Ionization Rate in the SHE–Boltzmann MOSFET Reproduction

*Resolved — the controlled trio is genuinely non-monotone; Richardson suppressed*

Reproduction study, follow-up to Revision 5 (code commit 676e74f+dirty; Python 3.13.15, NumPy 2.5.1)

August 21, 2026

## Abstract

Revision 5 established that Anderson acceleration converges the coupled SHE↔Poisson↔hole fixed point on each fixed mesh, but that the tail-sensitive impact-ionization rate  $G_{ii,\max}$  remained *spatially* mesh-dependent. This interim note documents the controlled spatial-convergence campaign that followed, now complete. Five results are firm: (i) the earlier uniform  $40 \times 34 \rightarrow 60 \times 50 \rightarrow 80 \times 66$  sequence was itself *under-resolving* the drain avalanche, so its values were inflated; (ii) a single further drain-refined mesh cannot settle the question because refining resolution and changing the drain grading law are confounded; (iii) a properly controlled self-similar trio  $71 \times 58 \rightarrow 100 \times 70 \rightarrow 141 \times 85$  — fixed grading law, byte-identical frozen energy grid (SHA1-verified),  $\sqrt{2}$  interval refinement — is the first sequence capable of answering grid stability cleanly; (iv) the nominal 60-outer-iteration budget *materially* truncated the nonlinear settling on the finer meshes, so node values must be accepted on an observable-plateau gate, not an iteration count; and (v) with all three nodes equalized to  $\sim 10^{-4}$  relative plateau quality, the controlled trio is **genuinely non-monotone** ( $2.990 \rightarrow 2.135 \rightarrow 2.413 \times 10^{25} \text{ cm}^{-3}\text{s}^{-1}$ ): the down-then-up reversal exceeds the combined plateau band by a factor  $\sim 870$ , so *no* Richardson continuum extrapolation is admissible. The drain-localized  $G_{ii,\max}$  is therefore *not* in an asymptotic spatial-convergence regime along this refinement path; the model gap to Liang’s top contour remains order  $10^2$  and is stated only as a not-converged tested-grid quantity.

**Live status. Resolved.** All three controlled nodes are converged to comparable ( $\sim 10^{-4}$ ) plateau quality. The trio is **genuinely non-monotone**; the reversal is numerically unambiguous ( $S_{\text{rev}} \approx 870$ ), so **Richardson extrapolation is suppressed** and no continuum  $G_{ii,\max}$  is quoted from this trio.

## 1 Scope and the three-way convergence distinction

The quantity under study is the peak impact-ionization generation rate  $G_{ii,\max}$  (and its cross-sectional integral  $G_{\text{tot}} = \int G_{ii} dA$ ) at the drain of the  $0.35 \mu\text{m}$  LDD NMOS at  $V_{gs} = V_{ds} = 3 \text{ V}$ ,  $V_{th} = 0.614 \text{ V}$ . Three distinct notions of “convergence” must be kept separate:

- **Coupling (fixed-point) convergence** — does the nonlinear outer loop reach its fixed point on a *given* mesh? *Resolved* in Rev. 5 by Anderson acceleration.
- **Spatial (mesh) convergence** — does the plateaued node value stop moving as the real-space mesh is refined? *The subject of this note.*

- **Energy-grid sensitivity** — dependence on  $\Delta H$ ,  $\epsilon_{\max}$ , and the frozen  $H$ -domain. Small ( $\sim 2.4\%$  one-way  $\Delta H$  in Rev. 5); here energy-grid variation is *removed as a between-mesh confounder* by construction (one common frozen  $H$ -grid across all meshes) — which does not eliminate energy-discretization error itself, only its variation across the trio.

## 2 The uniform sequence was under-resolved

The uniform-mesh drain sequence used in Rev. 5 is:

| grid           | drain $dx$ ( $\mu\text{m}$ ) | $G_{ii,\max}$ ( $\text{cm}^{-3}\text{s}^{-1}$ ) | $G_{\text{tot}}$ ( $\text{cm}^{-1}\text{s}^{-1}$ ) |
|----------------|------------------------------|-------------------------------------------------|----------------------------------------------------|
| $40 \times 34$ | 0.020                        | $8.04 \times 10^{26}$                           | $3.10 \times 10^{15}$                              |
| $60 \times 50$ | 0.013                        | $2.20 \times 10^{26}$                           | $8.27 \times 10^{14}$                              |
| $80 \times 66$ | 0.0101                       | $1.54 \times 10^{26}$                           | $5.81 \times 10^{14}$                              |

$G_{ii,\max}$  and  $G_{\text{tot}}$  fall by the *same* factors ( $-73\%$  then  $-30\%$ ), so the mesh dependence is in the total avalanche magnitude, not merely peak-cell placement: coarse grids over-populate the high-energy tail at the under-resolved drain hotspot. A drain-graded  $100 \times 70$  mesh (drain  $dx = 0.0027 \mu\text{m}$ ,  $\sim 3.8\times$  finer at the drain) plateaus at  $G_{ii,\max} = 1.80 \times 10^{25}$  — roughly an order of magnitude below the “finest uniform”  $80 \times 66$  — confirming the uniform values were inflated by drain under-resolution.

## 3 Why one further mesh is not enough: a confounded fifth point

A second drain-refined mesh ( $140 \times 84$ , drain  $dx = 0.00126 \mu\text{m}$ ) plateaus cleanly at  $G_{ii,\max} = 2.72 \times 10^{25}$  — *above* the  $100 \times 70$  value. Taken at face value this looks non-monotone, but that mesh changed *two* things at once: it refined resolution *and* altered the drain grading (ratio  $6 \rightarrow 10$ , width  $W = 0.09 \rightarrow 0.06 \mu\text{m}$ ). The  $+51\%$  change cannot be attributed to resolution alone.

**Honesty note.** This is exactly why a heterogeneous refinement sequence cannot establish spatial convergence: resolution and node-distribution effects are entangled. Richardson extrapolation on such a sequence would be meaningless. The remedy is a *self-similar* family in which only  $N_x, N_y$  change.

## 4 The controlled self-similar trio

We fix the grading law (ratio = 6,  $W = 0.09 \mu\text{m}$ ,  $X_C = 0.60 \mu\text{m}$ ;  $y = L_y s^{1.8}$ ) and refine only resolution, choosing node counts so that the *measured* drain  $dx$  and surface  $dy$  both refine by  $\approx \sqrt{2}$ :

| node  | $N_x \times N_y$ | drain $dx$ ( $\mu\text{m}$ ) | surface $dy$ ( $\mu\text{m}$ ) | $r_x$  | $r_{y,\text{surf}}$ |
|-------|------------------|------------------------------|--------------------------------|--------|---------------------|
| $G_1$ | $71 \times 58$   | 0.00380                      | 0.000484                       | —      | —                   |
| $G_2$ | $100 \times 70$  | 0.00269                      | 0.000343                       | 1.4144 | 1.410               |
| $G_3$ | $141 \times 85$  | 0.00190                      | 0.000241                       | 1.4144 | 1.425               |

The middle node coincides with the drain-refined  $100 \times 70$  mesh, anchoring the sequence.

### Comparability guards (enforced by the driver, fail-fast).

- **One common frozen energy grid.** A single imposed span (COMMON\_PHI\_SPAN =  $[-0.60, 3.65]$  V, bounding the measured  $\phi \in [-0.348, 3.586]$  V) fixes the  $H$ -grid so it is *byte-identical* across all meshes; the realized  $H$ -array is SHA1-hashed (b4319bb03303 on all nodes) and the extrapolation script refuses to run if the hashes differ. A converged  $\phi$  leaving the span is a hard abort. This removes any smuggled energy-discretization change from a spatial-only study.
- **Identical solver settings** —  $\Delta H = 12.5$  meV,  $\epsilon_{\max} = 3.02$  eV, `she_tol` =  $10^{-8}$ , Anderson depth 6,  $\beta = 0.5$ , step-cap 5, restart 6, same stopping rule and last-4 plateau definition — printed as a run manifest before every solve.
- **Measured spacings recorded** (drain  $dx$ , surface  $dy$ , and  $dy$  at the avalanche depth), with a warning that the depth spacing refines at  $\approx 1.2$ , *not*  $\sqrt{2}$ : the sequence is surface-isotropic (appropriate for the surface-localized  $G_{ii,\max}$ ) but not globally isotropic (relevant for the depth-sampling  $G_{\text{tot}}$ ).

## 5 Methodological finding: iteration budget vs. plateau

The finest node exposed a systematic issue. At the nominal 60-iteration cap,  $141 \times 85$  read  $G_{ii,\max} \approx 2.20 \times 10^{25}$  with a still-drifting observable; continued under *identical* settings it climbed and locked at  $2.413 \times 10^{25}$  (last-4 spread  $6 \times 10^{-5}$ ). Likewise  $71 \times 58$  moved from 3.069 to  $2.990 \times 10^{25}$  on continuation.

**Honesty note.** Equal outer-iteration counts do *not* imply equal fixed-point convergence quality across meshes: the finer mesh needs substantially more nonlinear iterations to settle. Node values must therefore be accepted on an *observable-plateau* gate —  $R_\phi$  at its floor *and* last-4 spreads at the  $O(10^{-4})$  level ( $\lesssim 3 \times 10^{-4}$ ) in both  $G_{ii,\max}$  and  $G_{\text{tot}}$ , applied identically to all three nodes — not on a fixed iteration budget. Operationally, “ $R_\phi$  at its floor” means the true fixed-point residual  $\max |G(\phi) - \phi|$  has stopped descending and merely oscillates within a noise band set by the inner solve (`she_tol` =  $10^{-8}$ ): here  $R_\phi \lesssim 10^{-3}$  V with no monotone decrease over the last several iterations, rather than a value still trending downward. (The final  $R_\phi$  never reaches the nominal  $10^{-4}$  V `tol_phi`; that tolerance is not the operative gate — the observable plateau is.) This is itself a reportable numerical lesson, and it is why the coarser nodes are re-flattened before any differencing.

## 6 Frozen nodes (all equalized)

| node                      | drain $dx$<br>( $\mu\text{m}$ ) | $G_{ii,\max}$<br>( $\text{cm}^{-3}\text{s}^{-1}$ ) | spread<br>( $G_{ii,\max}$ ) | $G_{\text{tot}}$<br>( $\text{cm}^{-1}\text{s}^{-1}$ ) | spread<br>( $G_{\text{tot}}$ ) | $R_\phi$<br>(V)      |
|---------------------------|---------------------------------|----------------------------------------------------|-----------------------------|-------------------------------------------------------|--------------------------------|----------------------|
| $G_1$ ( $71 \times 58$ )  | 0.003803                        | $2.990 \times 10^{25}$                             | $1.5 \times 10^{-4}$        | $8.983 \times 10^{13}$                                | $2.3 \times 10^{-4}$           | $8.7 \times 10^{-5}$ |
| $G_2$ ( $100 \times 70$ ) | 0.002689                        | $2.135 \times 10^{25}$                             | $2.9 \times 10^{-4}$        | $6.027 \times 10^{13}$                                | $2.2 \times 10^{-4}$           | $5.8 \times 10^{-4}$ |
| $G_3$ ( $141 \times 85$ ) | 0.001901                        | $2.413 \times 10^{25}$                             | $6.1 \times 10^{-5}$        | $6.593 \times 10^{13}$                                | $5.2 \times 10^{-5}$           | $1.8 \times 10^{-4}$ |

Both observables carry an independent plateau gate.  $G_{\text{tot}}$  plateau half-ranges are  $u_i^{G_{\text{tot}}} = \{1.02, 0.66, 0.17\} \times 10^{10} \text{ cm}^{-1}\text{s}^{-1}$ ; combined  $\sqrt{(u_2^{G_{\text{tot}}})^2 + (u_3^{G_{\text{tot}}})^2} = 6.84 \times 10^9$ .

All last-4 spreads — in *both*  $G_{ii,\max}$  ( $6.1 \times 10^{-5}$ ,  $1.5 \times 10^{-4}$ ,  $2.9 \times 10^{-4}$ ) and  $G_{\text{tot}}$  ( $2.3 \times 10^{-4}$ ,  $2.2 \times 10^{-4}$ ,  $5.2 \times 10^{-5}$ ) — are at the  $O(10^{-4})$  level (gate  $\lesssim 3 \times 10^{-4}$ , applied identically to both observables on all three nodes), so every node passes the same plateau gate ( $G_{ii,\max}$  plateau bands  $u_i \sim 10^{21}$ ). The equalization was essential and consequential: on continuation  $G_2$  rose from its 60-iteration value 1.806 to  $2.135 \times 10^{25}$  (+18%),  $G_1$  fell  $3.069 \rightarrow 2.990$ , and  $G_3$  rose  $2.20 \rightarrow 2.413$ . Comparing the equalized fixed points — not the truncated ones — is what makes the verdict below defensible.

## 7 Analysis framework and applied verdict

The evaluation order was fixed *before*  $G_2$  converged, and was not changed by the outcome:

1. **Monotonicity with numerical significance.** With signed corrections  $\Delta_{12} = G_1 - G_2$ ,  $\Delta_{23} = G_2 - G_3$  and per-node plateau bands  $u_i$  (last-4 half-range), a sign reversal is judged by the *heuristic* ratio

$$S_{\text{rev}} = \frac{|\Delta_{\text{reversal}}|}{\sqrt{u_i^2 + u_j^2}} \quad (\text{reversal-to-plateau-uncertainty ratio, not a statistical } \sigma).$$

$S_{\text{rev}} \gg 1$ : the reversal far exceeds residual fixed-point drift  $\Rightarrow$  genuinely non-monotone  $\Rightarrow$  suppress extrapolation.  $S_{\text{rev}} \lesssim O(1)$ : unresolved by plateau precision  $\Rightarrow$  read as grid stability near the finest nodes.

2. **Contraction (only if monotone).** Require  $|\Delta_{23}| < |\Delta_{12}|$ .
3. **Richardson (only if contracting).** With near-constant ratio  $r \approx \sqrt{2}$ ,

$$p = \frac{\ln |\Delta_{12}/\Delta_{23}|}{\ln r}, \quad G_\infty = G_3 + \frac{G_3 - G_2}{r^p - 1},$$

computed by an authoritative unequal-ratio 3-point solve on the full-precision measured spacings (drain- $dx$  ratios 1.41436, 1.41437, i.e.  $\sqrt{2}$  to  $4 \times 10^{-4}$ ) with the  $\sqrt{2}$  closed form as a transparency cross-check. Because only three controlled meshes exist,  $G_\infty$  is reported *as a bundle* with  $p$ , the correction ratio  $|\Delta_{23}|/|\Delta_{12}|$ , and the two- $h$  agreement — as evidence of asymptotic behaviour, not proof of the continuum value.

**Applied verdict.** The converged trio is

$$G_1 = 2.990 \times 10^{25}, \quad G_2 = 2.135 \times 10^{25}, \quad G_3 = 2.413 \times 10^{25} \text{ cm}^{-3}\text{s}^{-1},$$

with signed corrections  $\Delta_{12} = +8.55 \times 10^{24}$  (decrease) and  $\Delta_{23} = -2.78 \times 10^{24}$  (increase): **down then up**. The middle node lies below *both* neighbours, so the sequence is non-monotone. The reversal magnitude relative to the plateau band is

$$S_{\text{rev}} = \frac{|\Delta_{23}|}{\sqrt{u_2^2 + u_3^2}} = \frac{2.78 \times 10^{24}}{3.2 \times 10^{21}} \approx 874 \gg 1,$$

i.e. the upturn exceeds the residual fixed-point drift by nearly three orders of magnitude. The integral  $G_{\text{tot}}$  behaves identically: with  $G_2^{G_{\text{tot}}} = 6.027 \times 10^{13}$ ,  $G_3^{G_{\text{tot}}} = 6.593 \times 10^{13}$  ( $|\Delta_{23}^{G_{\text{tot}}}| = 5.65 \times 10^{12}$ ) and combined plateau band  $6.84 \times 10^9$  (from the  $G_{\text{tot}}$  half-ranges in §6),  $S_{\text{rev}}^{G_{\text{tot}}} = 5.65 \times 10^{12}/6.84 \times 10^9 \approx 826$ . All comparability gates pass: mesh-law/solver signatures identical, frozen  $H$ -array SHA1 **b4319bb03303** identical on all three, and (full-precision measured spacings)  $r_x = 1.41436$ ,  $1.41437$ ,  $r_{y,\text{surf}} = 1.410$ ,  $1.425$ .

**Honesty note.** The controlled trio is *genuinely non-monotone*, not merely unconverged. Because the non-monotonicity survived equalizing all three fixed points to  $\sim 10^{-4}$  quality, it cannot be dismissed as unequal convergence quality. This sign-changing trio does not support the assumed leading-order single-power Richardson model, so **extrapolation is suppressed**: no continuum  $G_{ii,\text{max}}$  is inferred from this trio. This is the defensible ceiling of the spatial study along this refinement path.

## 8 Model gap (a not-converged tested-grid statement)

Across the controlled trio the converged nodes span  $G_{ii,\text{max}} = 2.135\text{--}2.990 \times 10^{25} \text{ cm}^{-3}\text{s}^{-1}$ ; the finest mesh gives  $2.413 \times 10^{25}$ . Relative to Liang's top published  $G_{ii}$  contour ( $2.3 \times 10^{27} \text{ cm}^{-3}\text{s}^{-1}$ ) this is a factor 77–108, i.e. order  $10^2$ , below.

**Honesty note.** Because the controlled sequence is non-monotone, no converged spatial value and no continuum extrapolation can be *inferred from this trio* to anchor a final model gap. The order- $10^2$  figure is a *tested-grid* statement over the drain refinements performed, not a continuum/model gap. The remaining discrepancy is *consistent with a substantial model/physics discrepancy after the tested spatial refinements* (surrogate Keldysh impact-ionization rate on reconstructed doping, as in Rev. 5), but it cannot yet be called definitively physics-dominated because the spatial error is not bounded by a continuum study. What the present study does show firmly is that the *numerical* spatial contribution is larger and more subtle than a smooth monotone convergence would imply.

## 9 Conclusions and outlook

- **Coupling convergence:** resolved (Anderson, per mesh).
- **Spatial convergence: not demonstrated.** The controlled three-grid sequence is *significantly* non-monotone ( $2.990 \rightarrow 2.135 \rightarrow 2.413 \times 10^{25}$ ,  $S_{\text{rev}} \approx 874$ ), so no continuum value can be inferred from this refinement trio. This does *not* establish that a continuum limit fails to exist — finer self-similar meshes could still enter a monotone asymptotic regime; it establishes that this trio has not entered one. The non-monotonicity itself is a firm, numerically-resolved result.
- **Energy grid:** energy-grid *variation* removed as a between-mesh confounder (one common frozen  $H$ -grid, SHA1-identical) — not a claim that energy-discretization error is zero.
- **Model gap:** order  $10^2$  below Liang’s top contour ( $77\text{--}108\times$ ), stated as a not-converged tested-grid quantity; consistent with a substantial model/physics discrepancy after the tested refinements, but not yet definitively physics-dominated (spatial error still unbounded).

**What would be needed to go further.** Because  $G_{ii,\text{max}}$  is a tail-sensitive moment at a near-singular drain hotspot, the natural next instruments are (a) a *second* controlled family — fixed  $y$ -mesh with systematic drain- $x$  refinement — to separate  $x$ - and  $y$ -discretization error and test whether non-monotonicity is direction-specific; and (b) a mesh law with finite surface slope so all local  $dy$  refine linearly, enabling a genuinely isotropic global study. Neither is required to state the present conclusion; both would sharpen *why* the drain avalanche resists monotone spatial convergence. Physics-model changes (surrogate impact-ionization rate, reconstructed doping) remain the dominant lever on the absolute gap to Liang, and are deferred by design until the numerical characterization is complete.

**Honesty note.** Methodological lesson to carry forward: a fixed outer-iteration budget does not guarantee comparable fixed-point quality across meshes. Here the coarser nodes had to be continued well beyond 60 iterations, and their converged values shifted by up to +18%; the non-monotonicity verdict is only trustworthy *because* it was evaluated on equalized, plateau-gated fixed points with an explicit reversal-significance test ( $S_{\text{rev}}$ ), not on truncated iterates.